OJT – Project Proposal (PRD)

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Track: Product Development
Project Title: Loci - An Audio-First Spatial Guide for Structured B2B Environments

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1. Problem Understanding

1.1 What is the problem statement in your own words?

In large structured indoor environments such as campuses, museums, and institutional buildings, users frequently move through meaningful spaces without understanding their context or significance.

Existing solutions rely heavily on:

- visual interfaces (maps, screens, signage), or
- precise real-time positioning systems

Both approaches fail in practice:

- Visual systems interrupt natural movement and increase cognitive load
- Indoor positioning systems are unreliable, leading to incorrect or mistimed guidance

There is no system that:

- works seamlessly while users are in motion
- minimizes interaction
- guarantees correctness over frequency

1.2 Why does this problem exist or matter? (WHY Component)

Most navigation systems are designed with a **precision-first assumption**, expecting continuous accurate location data.

In indoor environments:

- GPS is unreliable
- WiFi signals are inconsistent
- Sensor noise introduces ambiguity

As a result:

- Systems either guess → causing incorrect guidance
- Or overuse UI → increasing user effort

In structured B2B environments:

- incorrect information reduces trust
- noisy systems degrade user experience

This project adopts a different philosophy:

> **“Correctness over coverage. Silence over misinformation.”**

1.3 Key inputs and expected outputs

Inputs	Processing	Outputs
User movement (IMU, GPS signals)	Deterministic spatial reasoning	Contextual audio narration
Predefined spatial graph (zones, paths)	Confidence evaluation	Correctly timed guidance
Structured knowledge base	Bounded AI retrieval	Grounded responses
Voice queries	Speech-to-text + intent extraction	Context-aware answers
Sensor signals (optional WiFi fallback)	State machine orchestration	Silence when uncertain

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2. Functional Scope

2.1 What are the core features you plan to build (must-haves)?

- Audio-first spatial guidance (no dependency on screens)
- Indoor-only navigation within structured environments
- Deterministic zone sequencing using predefined graph
- Confidence-based narration triggering
- Silence enforcement when confidence is low
- Event-driven voice interaction pipeline
- Speech-to-text → AI → text-to-speech loop
- Controlled AI Q&A (bounded to approved knowledge)
- Prevention of overlapping or conflicting audio playback
- Manual + guided triggering for demo and fallback scenarios

- Minimal UI for system state and interaction

2.2 What stretch goals could you attempt if time permits?

- Improved indoor localization via sensor fusion
- Adaptive confidence scoring using historical signals
- Multi-floor navigation support
- Dynamic route selection (non-linear exploration)
- Real-time analytics for system evaluation

2.3 Which libraries or tools will you use?

- **Frontend:** React Native (Expo), TypeScript
- **Backend:** Node.js, Express
- **Database:** PostgreSQL (Neon)
- **Authentication:** JWT, bcrypt
- **Speech-to-Text:** Deepgram (nova-2)
- **Text-to-Speech:** Expo Speech
- **AI Layer:** Groq LLM (LLaMA-based)
- **Sensors:** Accelerometer (IMU), GPS, optional WiFi fallback
- **Networking:** REST APIs via ngrok (development)

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3. System & Design Thinking

3.1 Sketch or describe your app flow / pipeline

User starts guided session

→ Spatial orchestrator initializes at entry state

→ Sensor signals continuously evaluated (motion, GPS, optional WiFi)

→ Deterministic zone progression evaluated

→ Confidence threshold computed

→ If confidence high → trigger narration

→ If confidence low → maintain silence

→ User may interrupt via voice query

→ Audio recorded → transcribed (Deepgram)

→ AI retrieves bounded answer → converted to speech

→ Playback managed with audio session control

Flow Summary:
Structured data → confidence reasoning → intentional audio → silence when unsure

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3.2 Core Architectural Concepts

1. Spatial Orchestrator (Finite State Machine)

- Controls navigation flow
- Maintains states such as:
  - OUTDOOR (removed in implementation)
  - ENTRANCE_LOCKED
  - NAVIGATING
  - NARRATING
  - PAUSED
- Ensures deterministic transitions

2. EventBus (Pub-Sub System)

- Decouples components
- Enables asynchronous communication between:
  - Navigation layer
  - Speech engine
  - UI

3. Speech Engine Pipeline

Mic Input → expo-av

→ Deepgram (STT)

→ Backend AI (Groq)

→ Response Text

→ expo-speech (TTS)

4. Confidence Layer

- Combines:
  - motion state (walking/still)
  - GPS approximation
  - optional WiFi signals
- Outputs:

    - High → trigger narration
    - Low → silence

5. Audio Control System

- Prevents overlap between:
    - narration
    - recording
    - AI responses
- Uses mutex guards and session resets

6. Data Layer

Structured JSON-based system:

- `graph.json` → spatial connectivity
- `knowledge.json` → AI content
- `audio_index.json` → narration mapping
- `wifi_fingerprints.json` → optional localization

3.3 Algorithms & Logic

- Graph traversal for zone progression
- Confidence threshold gating
- Intent extraction from voice input
- Keyword + zone-based retrieval
- Fallback logic when AI fails

3.4 Testing Strategy

- Controlled walkthrough testing
- Repeated runs for consistency
- Audio correctness validation
- Sensor failure simulation
- Voice pipeline latency checks

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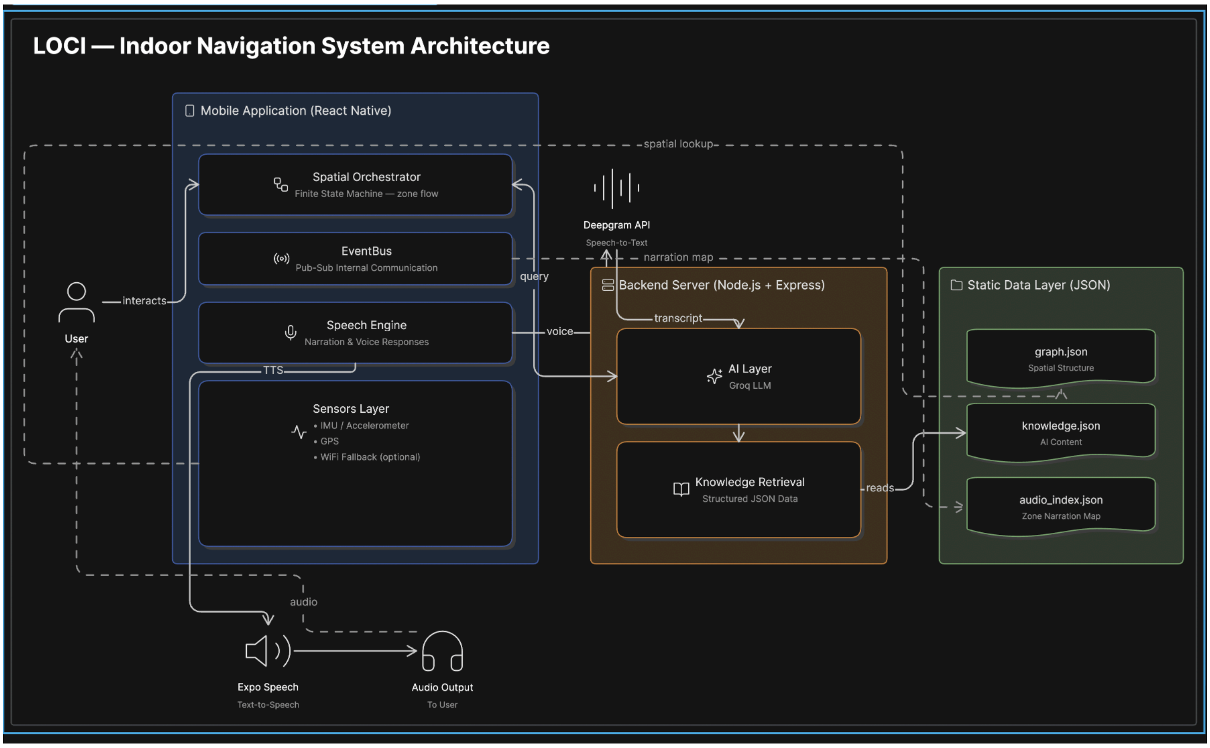
4. High Level Design (HLD)

LOCI follows a modular, event-driven architecture centered on a React Native mobile app. The app includes a Spatial Orchestrator (finite state machine) for deterministic navigation, an EventBus for decoupled communication, a Speech Engine for audio interaction, and a Sensors Layer (IMU, GPS, optional WiFi) for context.

User speech is recorded via expo-av, transcribed using Deepgram, processed by a bounded AI layer (Groq LLM) on the backend, and returned as audio using expo-speech.

The backend (Node.js + Express) handles API routing, controlled AI inference, and structured knowledge retrieval from JSON data (graph, knowledge, audio index).

The system prioritizes deterministic navigation, using AI only for language tasks, ensuring reliable and predictable behavior.



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5. Timeline & Milestones (12 Weeks)

Phase	Duration	Deliverables
Planning & Design	Week 1–2	PRD, HLD, system architecture
Data Modeling	Week 3–4	Spatial graph + knowledge design
Mobile App Core	Week 5–6	UI, routing, navigation base

Orchestrator & Audio	Week 7–8	FSM + narration system
Voice & AI Integration	Week 9–10	STT + AI + TTS pipeline
Testing & Demo	Week 11–12	End-to-end validation

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6. Risks & Dependencies

- Indoor localization without precise positioning
- Managing audio conflicts (recording vs playback)
- Ensuring deterministic behavior under noisy inputs
- Maintaining strict AI boundaries

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7. Evaluation Readiness

7.1 Demonstration Plan

- Guided indoor walkthrough
- Manual zone triggering demo
- Voice interaction demo
- End-to-end narration pipeline

7.2 Success Metrics

- Zero incorrect audio playback
- Stable zone progression
- Clear, timely narration
- Reliable fallback responses
- Minimal user interaction required

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